

Isaac Pressman

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EDUCATION

California Polytechnic State University, San Luis Obispo (Cal Poly) San Luis Obispo, CA
Bachelor of Science in Computer Science – GPA 3.94 Sep. 2024 – May 2028
Relevant Coursework: Data Structures, Computer Organization, Object Oriented Programming, Systems Programming, Algorithms, Intro to Computer Security, Database Systems, Intro to Data Science, Operating Systems

EXPERIENCE

Software Engineer Intern, AI Solutions Sep. 2026 – Present
Cloudwick Remote (Part-time)

- Designed stale-page reconciliation for a RAG ingestion pipeline, distinguishing confirmed webpage deletions from transient crawl failures before removing stale content from the index and preventing citations to deleted sources.
- Redesigned **AWS WAF integration** for custom API domains, identifying request paths that bypassed existing protections and implementing CloudFormation changes to preserve API security.

Software Engineer Intern June 2026 – Aug. 2026
Coherent Corp. Fremont, CA

- Built a **Python stress-testing framework** exercising thousands of I2C operations across bus speeds and I/O timing conditions, cutting manual OSFP firmware validation from **3 hours to 30 minutes**.
- Designed and implemented **Python automation** for previously unexercised firmware paths after auditing existing test suites, expanding firmware coverage by **8%**.
- Diagnosed firmware defects in lane-control and CMIS-compliance behavior by correlating automated test results with device logs and partnering with firmware engineers to isolate root causes before release.

Software Engineer Intern Jan. 2026 – June 2026
Teradyne Agoura Hills, CA

- Engineered a **gRPC bridge** across incompatible .NET 4.6 and .NET 10 runtimes, enabling a legacy C# system to control **Playwright automation** and power the full nightly CI smoke test suite.
- Architected an **LLM-driven framework** in C# that auto-generates firmware validation cases, increasing coverage from **85% to 95%** and uncovering 100+ driver and control-path edge cases.
- Refactored **15,000+ lines** of legacy C# system components to eliminate duplicated logic across hardware control paths, reducing code redundancy by **73%**.

Research Software Developer Aug. 2024 – Sep. 2025
UC Berkeley Optometry Berkeley, CA

- Developed a Python (OpenCV, NumPy) computer vision pipeline to measure tear film lipid thickness with **13nm** accuracy, matching **\$20k+** commercial measurement systems.
- Automated pupil/iris segmentation across **50K+ clinical video frames**, reducing analysis from **5 minutes to 10 seconds** per video while maintaining research-grade accuracy, enabling **30x faster** research throughput.

PROJECTS

The Shoe Assist | *FastAPI (Python), Next.js, PostgreSQL + pgvector, RAG, Docker, Kubernetes, Terraform*

- Built a full-stack basketball shoe recommendation platform with REST APIs and a **RAG engine** using Postgres, pgvector, and **vector embeddings** to personalize by playstyle, budget, and aesthetic.
- Provisioned Kubernetes infrastructure as code with Terraform; deployed containerized Next.js and FastAPI services with ingress, health probes, horizontal pod autoscaling, and zero-downtime rollouts via GitHub Actions CI/CD.

IT Pal | *Electron, JavaScript, Node.js, GPT-4o-mini (vision, via OpenRouter), PowerShell*

- Built an **accessible, voice-first desktop assistant** that turns seniors' plain-language requests into vision-grounded, on-screen automation with spoken confirmation in under 6 seconds.
- Won **2nd place and \$1,500** among 228 participants at AWS-sponsored KiroHacks Cal Poly

TECHNICAL SKILLS

Languages: Java, Python (OpenCV, NumPy, SciPy), C/C++, C#, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks: Django, .NET, React, React Native, Node.js, Express.js, FastAPI, Unity, Electron
Technologies: Kubernetes, Git, Docker, Terraform, AWS (S3, EC2, IAM), Playwright, gRPC, I2C, Linux/Unix, Bash